Baseline Vitamin D, Preeclampsia, and Early-Pregnancy Gene Expression

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1 Introduction

Low maternal vitamin D in early pregnancy has been linked to a higher risk of preeclampsia [1]. CYP24A1 encodes the enzyme that inactivates the active form of vitamin D, so I examined its expression together with baseline vitamin D and pregnancy outcome. I also examined UTS2 and RUNDC3A, the two most variable of the supplied genes, to compare expression patterns. This report is descriptive and does not test hypotheses.

2 Methods

2.1 Data source and sample size

The data come from the study by Mirzakhani et al. [1] and are available from the Gene Expression Omnibus under accession GSE85307 [2]. The course provided three files: covariate data for 157 participants, expression values for 500 genes measured with the Affymetrix Human Gene 1.0 ST Array, and a linker file mapping probe IDs to gene names. Blood samples were collected between 10 and 19 weeks of gestation. The initial sample size was 157 participants (110 normal pregnancies and 47 with preeclampsia).

2.2 Data processing and final sample size

All data wrangling was performed in R. I removed the text label from each covariate value, converted the continuous covariates to numbers, matched expression columns to participants by their GEO sample ID, and replaced probe IDs with gene names using the linker file. Every sample ID and probe ID matched, and no covariate or expression value was missing. I therefore removed no participants, and the final sample size was 157. I derived a categorical vitamin D group from baseline 25-hydroxyvitamin D using a cutoff of 30 ng/mL (<30 vs. ≥ 30 ng/mL).

2.3 Analyses

I inspected histograms of each continuous covariate before choosing summary statistics. Maternal age was approximately symmetric and is reported as mean (SD); vitamin D and BMI were right-skewed and are reported as median [Q1, Q3]. Categorical variables are reported as n (%). The summary table is stratified by pregnancy outcome.

For CYP24A1, UTS2, and RUNDC3A, I used one plotting function to produce a histogram of expression, a scatterplot of expression against baseline vitamin D with an unadjusted linear fit and 95% confidence band, and a boxplot of expression by pregnancy outcome and vitamin D group. For the heatmap, I selected the ten genes with the highest standard deviation across

participants and applied hierarchical clustering with Euclidean distance and complete linkage to rows and columns, without row standardization. As an additional plot, I used a dumbbell plot to compare the median expression of the same ten genes between the two outcome groups, with genes ordered by the difference in medians (preeclampsia minus normal). All analyses were descriptive; I did not perform hypothesis tests or estimate adjusted effects.

2.4 Software

Analyses were conducted in R version 4.6.1 [3]. Figures were made with ggplot2 version 4.0.3 [4] and pheatmap version 1.0.13 [5]. Tables were made with knitr version 1.52 [6] and kableExtra version 1.4.1 [7], and the analysis was compiled with R Markdown [8].

2.5 Code availability

All code is available in my public GitHub repository:
https://github.com/llliii0088979-a11y/QBS103_DataScience_Project.

2.6 Use of AI and other resources

This was my first time using R, so I made frequent errors while writing the code. I used an AI assistant (Claude) to help me debug errors, suggest code for parts of the analysis (such as the dumbbell plot), and draft and edit the wording of this report. I also watched YouTube tutorials to help me understand R and ggplot2 concepts. I chose the research question, genes, and covariates, reviewed and ran all of the code myself, and checked that the results and text matched the data.

3 Results

3.1 Participant characteristics

I found lower median baseline vitamin D (17.7 vs. 21.8 ng/mL) and higher median BMI (31.24 vs. 26.54 kg/m^2) in participants with preeclampsia than in participants with normal pregnancies, while mean maternal age was similar (25.78 vs. 26.63 years). A larger share of the preeclampsia group had vitamin D below 30 ng/mL (87.2% vs. 70.9%; Table 1).

Table 1: Participant characteristics by pregnancy outcome.

Variable	Overall ($n = 157$)	Normal ($n = 110$)	Preeclampsia ($n = 47$)
Vitamin D (ng/mL), median [Q1, Q3]	20.4 [13.4, 29.5]	21.8 [13.85, 32.08]	17.7 [12.55, 27.4]
Maternal age (years), mean (SD)	26.38 (5.04)	26.63 (5.08)	25.78 (4.95)
Maternal BMI (kg/m^2), median [Q1, Q3]	27.33 [22.9, 33.27]	26.54 [22.35, 30.97]	31.24 [24.62, 36.41]
Pregnancy outcome, n (%)			
Normal	110 (70.1)	110 (100)	0 (0)
Preeclampsia	47 (29.9)	0 (0)	47 (100)
Vitamin D group, n (%)			
<30 ng/mL	119 (75.8)	78 (70.9)	41 (87.2)
≥30 ng/mL	38 (24.2)	32 (29.1)	6 (12.8)

3.2 CYP24A1

I observed a single, narrow peak in CYP24A1 expression near 4.8, with a small upper tail (Figure 1). The fitted line between baseline vitamin D and CYP24A1 expression was nearly flat, so I did not

see a clear linear pattern in this sample (Figure 2). Expression distributions overlapped across all four outcome and vitamin D subgroups. I interpreted the lower median in the preeclampsia, ≥30 ng/mL subgroup cautiously because it included only six participants (Figure 3).

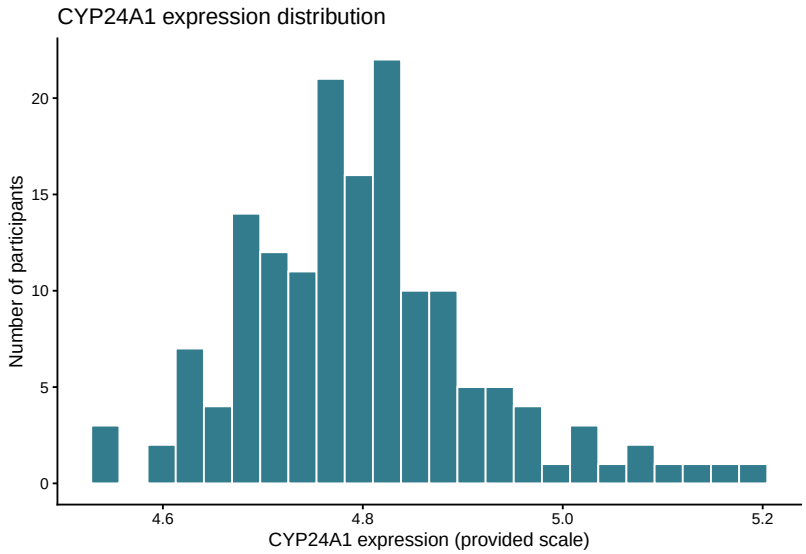


Figure 1: CYP24A1 expression in all 157 participants. The histogram uses 24 bins and the supplied expression scale.

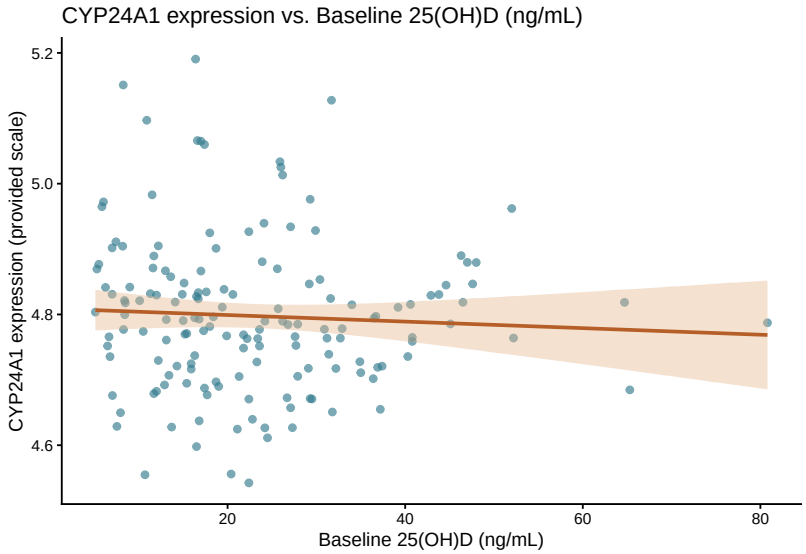


Figure 2: CYP24A1 expression and baseline vitamin D. The line is an unadjusted linear fit; shading shows the 95% confidence interval for the fitted mean.

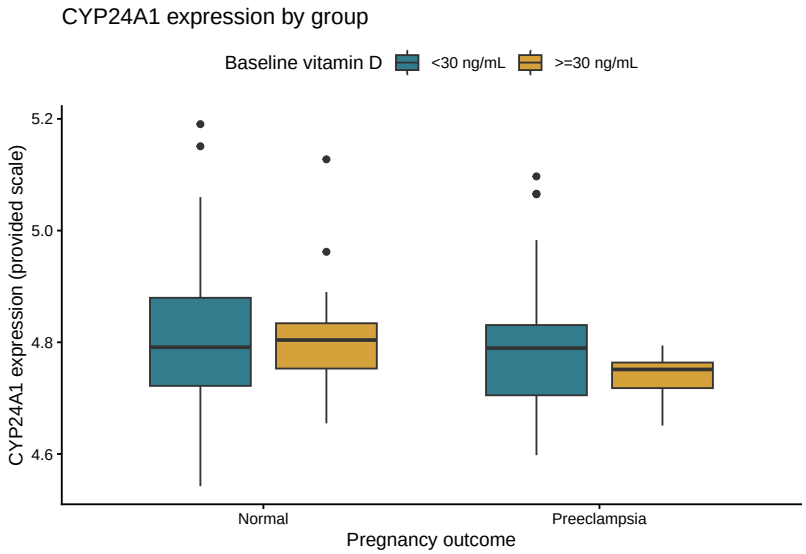


Figure 3: CYP24A1 expression by pregnancy outcome and vitamin D group. Boxes show medians and quartiles; whiskers extend to observations within 1.5 times the IQR, and points beyond them are outliers. Group sizes: <30 ng/mL, Normal 78 and Preeclampsia 41; ≥30 ng/mL, Normal 32 and Preeclampsia 6.

3.3 UTS2

UTS2 expression was clearly bimodal: most participants had values near 5, and a smaller group had values between about 6 and 8.5 (Figure 4). Both groups appeared across the range of baseline vitamin D, so the single linear fit passed between them and did not describe either group well (Figure 5). The six preeclampsia participants with vitamin D ≥30 ng/mL had a higher median UTS2 expression, but I did not treat this small subgroup as evidence of a difference (Figure 6).

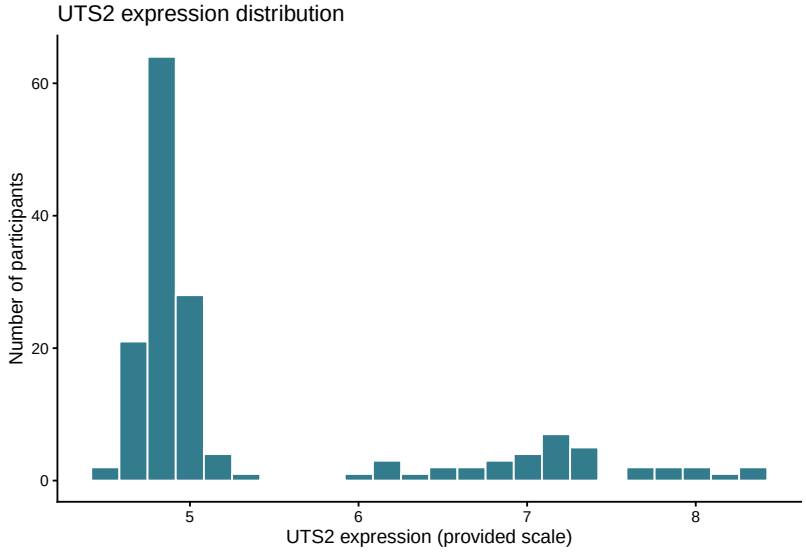


Figure 4: UTS2 expression in all 157 participants. The histogram uses 24 bins and the supplied expression scale.

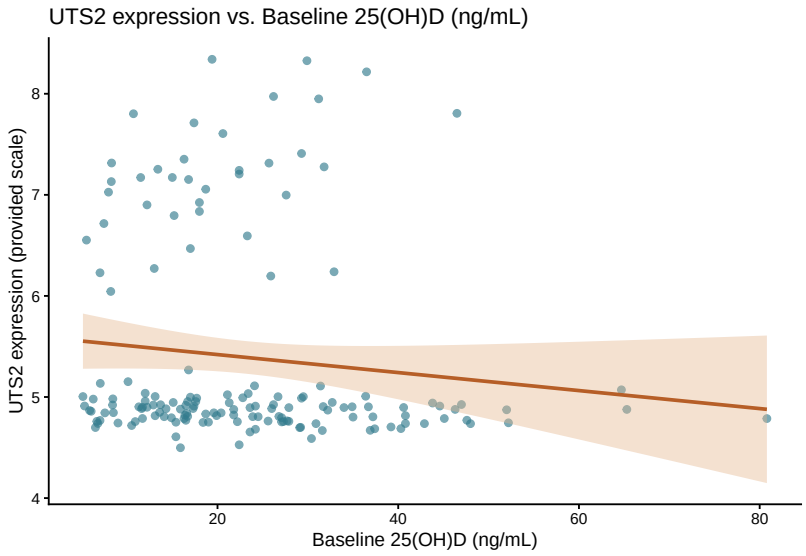


Figure 5: UTS2 expression and baseline vitamin D. The line is an unadjusted linear fit; shading shows the 95% confidence interval for the fitted mean.

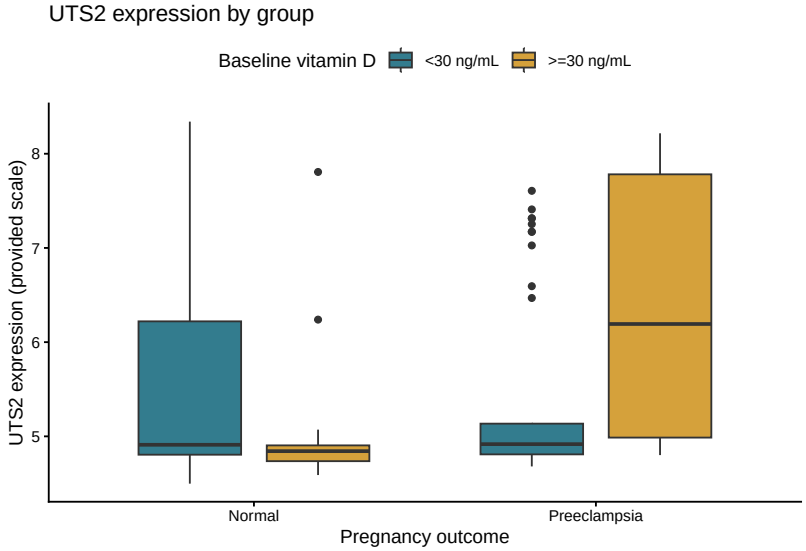


Figure 6: UTS2 expression by pregnancy outcome and vitamin D group. Boxes show medians and quartiles; whiskers extend to observations within 1.5 times the IQR, and points beyond them are outliers. Group sizes: <30 ng/mL, Normal 78 and Preeclampsia 41; ≥30 ng/mL, Normal 32 and Preeclampsia 6.

3.4 RUNDC3A

I observed a broader, roughly symmetric distribution of RUNDC3A expression centered near 9 (Figure 7). The fitted line sloped slightly downward with baseline vitamin D, with substantial scatter around it (Figure 8). Median RUNDC3A expression was lower in the preeclampsia group within both vitamin D groups, although the distributions overlapped (Figure 9).

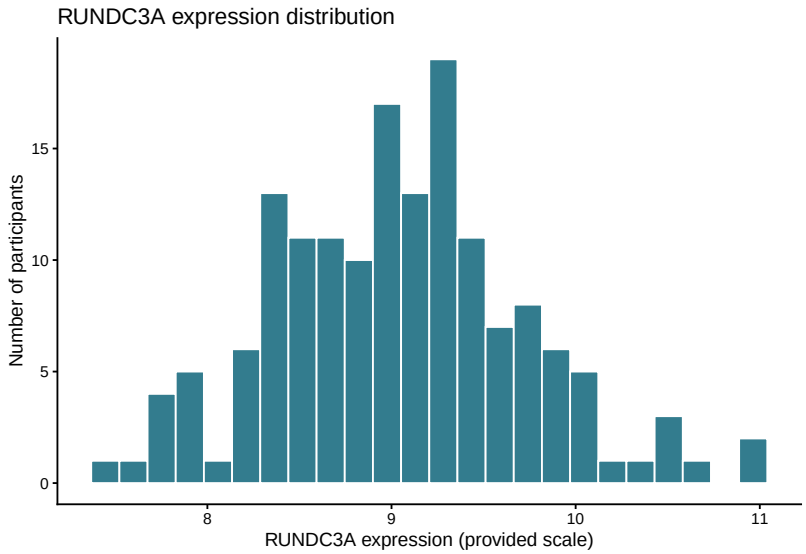


Figure 7: RUNDC3A expression in all 157 participants. The histogram uses 24 bins and the supplied expression scale.

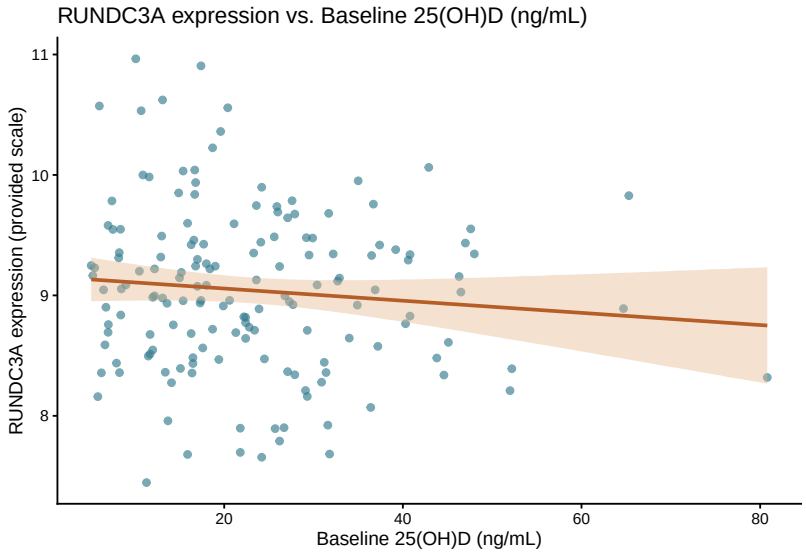


Figure 8: RUNDC3A expression and baseline vitamin D. The line is an unadjusted linear fit; shading shows the 95% confidence interval for the fitted mean.

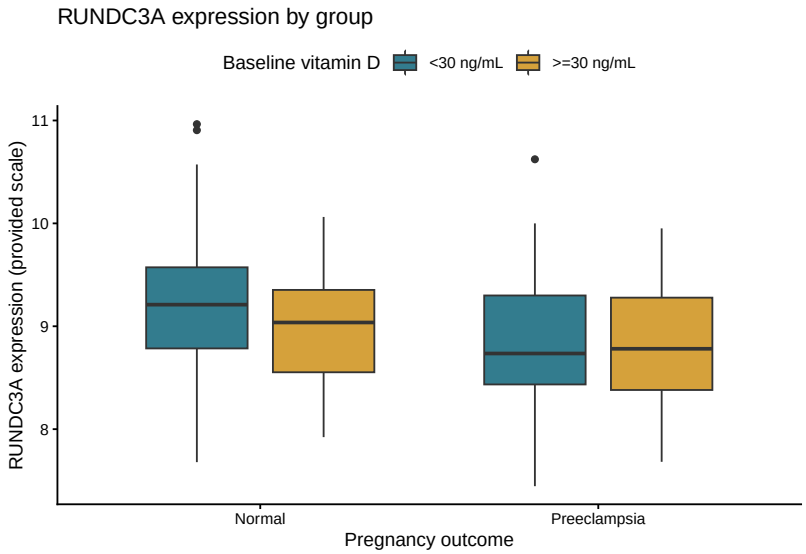


Figure 9: RUNDC3A expression by pregnancy outcome and vitamin D group. Boxes show medians and quartiles; whiskers extend to observations within 1.5 times the IQR, and points beyond them are outliers. Group sizes: <30 ng/mL, Normal 78 and Preeclampsia 41; ≥30 ng/mL, Normal 32 and Preeclampsia 6.

3.5 Ten-gene heatmap

Because I kept the supplied scale, the colors mainly reflected differences in overall expression level between genes, and the genes clustered into a lower-expression group (UTS2, CEACAM6, CLIC2, ZNF594) and a higher-expression group. One cluster of participants had higher UTS2 expression, matching the second UTS2 group in Figure 4. The tracking bars showed no visible separation by pregnancy outcome or vitamin D group (Figure 10). I treated the clustering as exploratory.

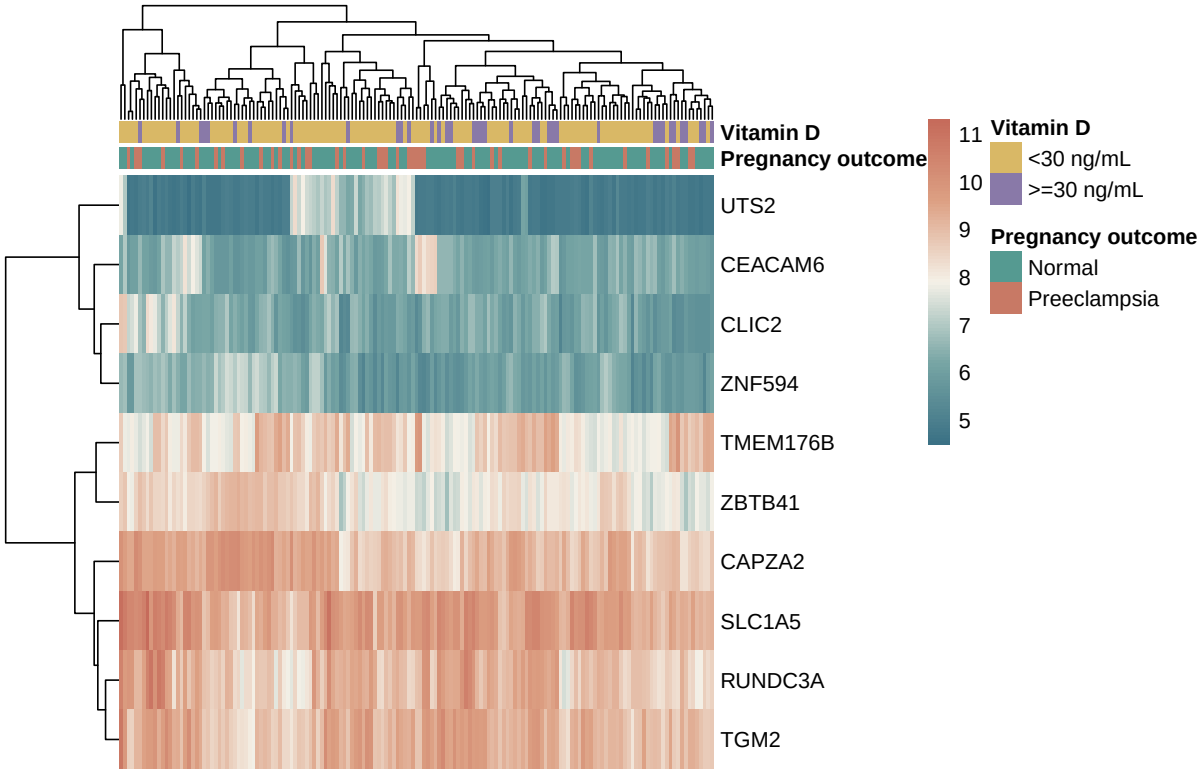


Figure 10: The ten genes with the highest standard deviation across 157 participants. Colors show expression on the supplied scale. Rows and columns use Euclidean distance and complete-linkage hierarchical clustering. Tracking bars show pregnancy outcome and baseline vitamin D group. Sample labels are hidden for readability.

3.6 Median expression of the ten heatmap genes by outcome

Among the ten most variable genes, I found the largest difference between outcome groups for RUNDC3A, with lower median expression in preeclampsia (8.74 vs. 9.15; difference −0.41), followed by TGM2 (−0.28), ZNF594 (−0.21), and CLIC2 (−0.19). The remaining six genes differed by 0.06 or less (Figure 11). I report these as unadjusted descriptive differences, not test results.

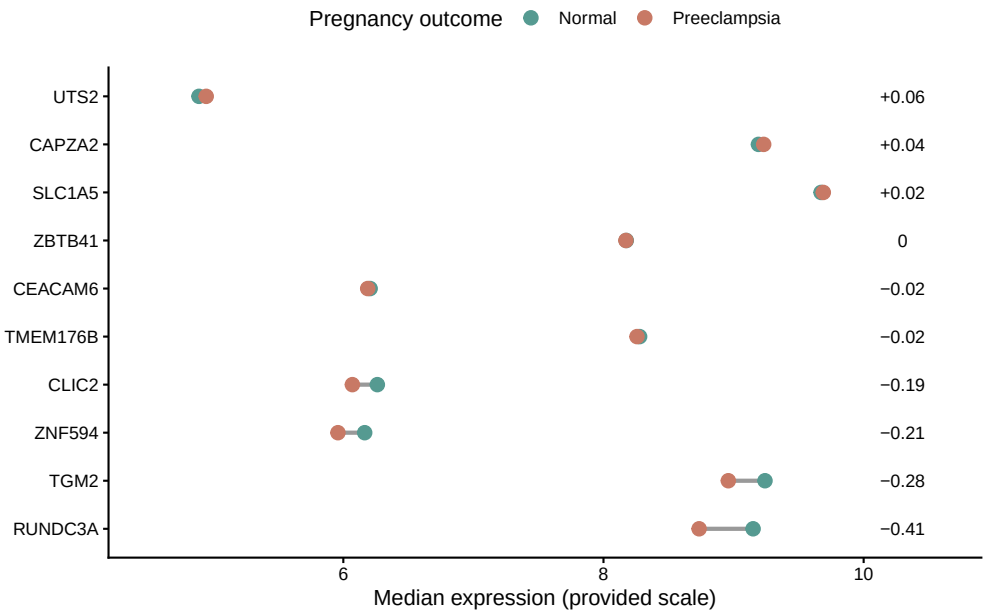


Figure 11: Median expression of the ten highest-variance genes in normal pregnancies ($n = 110$) and preeclampsia ($n = 47$). Each line joins the two group medians for one gene; numbers on the right give the difference (preeclampsia minus normal) on the supplied expression scale. Genes are ordered by this difference.

4 Discussion

In this sample, participants with preeclampsia had lower baseline vitamin D and were more often below 30 ng/mL, consistent with the direction reported by Mirzakhani et al. [1]. CYP24A1 expression showed no clear pattern with vitamin D or outcome in my analysis. Among the most variable genes, RUNDC3A had the largest outcome difference, consistent with its lower median in the preeclampsia group within both vitamin D groups. I consider these findings descriptive: the analyses are unadjusted, include no hypothesis tests, and some subgroups are small, especially the six preeclampsia participants with vitamin D $\geq$30 ng/mL.

References

[1] Hooman Mirzakhani, Augusto A. Litonjua, Thomas F. McElrath, et al. Early pregnancy vitamin D status and risk of preeclampsia. *Journal of Clinical Investigation*, 126(12):4702–4715, 2016. doi:10.1172/JCI89031.

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